Code:

```
abstract class EnergySource {
   String s_Id;
   String s_Name;
   double e_Gen;
   EnergySource(String s_Id, String s_Name, double e_Gen) {
      this.s_Id = s_Id;
      this.s_Name = s_Name;
      this.e_Gen = e_Gen;
   }
   abstract double calculateEfficiency();
   void displayDetails() {
      System.out.println("Source ID: " + s_Id);
      System.out.println("Source Name: " + s_Name);
      System.out.println("Energy Generated: " + e_Gen + " kWh");
      System.out.printf("Efficiency: %.2f\n",
calculateEfficiency());
      System.out.println("     ");
   }
}
class SolarEnergy extends EnergySource {
   SolarEnergy(String s_Id, String s_Name, double e_Gen) {
      super(s_Id, s_Name, e_Gen);
   }
   double calculateEfficiency() {
      return (e_Gen / 5000.0) * 100.0;
   }
}
class WindEnergy extends EnergySource {
   WindEnergy(String s_Id, String s_Name, double e_Gen) {
      super(s_Id, s_Name, e_Gen);
```

```
    }
    double calculateEfficiency() {
        return (e_Gen / 8000.0) * 100.0;
    }
}
public class EnergyManagementSystem {
    public static void main(String[] args) {
        EnergySource ref;
        ref = new SolarEnergy("SOL-101", "Solar Energy", 4200.0);
        ref.displayDetails();
        ref = new WindEnergy("WND-202", "Wind Energy Center", 5200.0);
        ref.displayDetails();
        System.out.println("Coded by: Kasham Adhikari");
System.out.println("Section: C");
System.out.println("USN: 1BF25CS122");
}
}
```

Output:

